

Suggested Teaching Guideline for
Public Key Infrastructure
PGCP-ITISS February 2026

Duration: 30 Theory hrs + 20 Lab hrs+ 10 Self-Learning hrs (Total: 60 hrs)

Objective: To introduce the student to PKI concepts.

Evaluation method: Theory exam – 40% weightage
Lab exam – 40% weightage
Internal Assessment – 20% weightage

List of Books / Other training material

Courseware: Cryptography & Network Security: Principles and Practices, Stallings William, (Pearson Education), 7th Edition

Reference:

- Cryptography & Network Security – Atul Kahate
- Understanding Digital Signatures: Establishing Trust Over the Internet and Other Networks – Gail Grant
- Windows Server 2008 PKI and Certificate Security [With CDROM] by Brian Kumar

Session 1: (2T)

Theory:

- Information Security
- Security Attacks & Threats

Session 2: (2T)

Theory:

- Basic Encryption Concepts
- File Encryption
- Encryption Folders (Graphical/ using cipher)

Session 3: (2T)

Theory:

- Cryptographic Fundamentals
- Cryptographic Ciphers (Symmetric and Asymmetric)
- Protocols (History, Usage, Key generation, Ciphering message)

Session 4: (2T)

Theory:

Symmetric Key Encryption

- DES
- AES, RC5

Asymmetric Key Encryption

- RSA

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- ECC

Session 5: (2T + 4L)

Theory:

- Diffie-Hellman Key Exchange
- Attacks against encryption
- Cryptographic issues

Lab Assignments:

Using Cryptool

- Symmetric and Asymmetric key encryption and decryption
- Symmetric key encryption such as Caesar, Vernam, DES, RC4, AES, Substitution, XOR, Triple DES, etc.
- Asymmetric key encryption using RSA, ECC

Session 6: (2T)

Theory:

Secure Hashing Methods

- SHA Secure Hash algorithm
- HMAC

Session 7: (2T+ 5SL)

Theory:

PKI Fundamentals

- Digital Signature
- Digital Certificate

Self Learning: [Note: SL based session 1-7]

- Explore about Information Security goals (CIA Triad) and threat mapping
- Difference between encryption, hashing, and digital signatures
- Find out Symmetric vs asymmetric encryption use cases
- Why Diffie-Hellman is required for secure key exchange
- Find out Secure hashing vs encryption (SHA vs AES)
- Role of HMAC in message integrity
- Common cryptographic implementation mistakes
- Study about Password Database Breach
 - ◆ Analyze How lack of salt leads to rainbow table attacks
 - ◆ Role of SHA and HMAC in secure password storage

Session 8: (2T)

Theory:

- CA
- Trust Model
- Certificate Issuance Process

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- Certificate Revocation (CRL, OCSP)
- Types and Classes of Certificate

Session 9: (2T + 4L)

Theory:

- Introduction to Aadhaar and e-Sign
- Time stamping Services

Lab:

Using XCA to create a Digital Signature

- Digitally Sign a Word document using the created certificate
- Digitally Sign a PDF document using the created certificate

Session 10: (2T + 4L)

Theory:

Public Key Cryptography Standards

- PKCS
- FIPS 140-2

Lab:

Using XCA to create a Digital certificate

- Create a digital certificate using XCA tool
 - Create CA using XCA first and then CA will issue certificate to website using XCA.
 - a. Use certificate for host website (<https://www.ditiss.local>)
 - b. Import it client browser to remove the self-signed certificate warning.

Session 11: (2T)

Theory:

- Strong Authentication
- Single Factor and Multi-factor authentication
- Single Sign-on Solutions
- Open-ID and OAuth
- Graphical Passwords

Session 12: (2T)

Theory:

- Authentication Protocols
- FIDO Authentication
- Zero Trust Architecture

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Session 13: (2T + 4L)

Theory:

Securing Websites and Emails

- SSL
- TLS
- PGP and S/MIME

Lab:

Using OpenSSL to create

- Creating self-signed Digital Certificates using OpenSSL.
- Creating digital certificate of Hierarchical Trust Model for Intranet
 - Setup a PKI infrastructure using openssl as following:

>> Root CA (on a Debian OS - rtca.pgditiss.local)

>> Sub CA (on a Debian OS - sbca.pgditiss.local)

Setup a HTTPS website (<https://www.pgditiss.local>) on Apache and access the same from the windows base machine.

Note:

- >> The Certificate for www.pgditiss.local to be issued by sbca.pgditiss.local
- >> Configure DNS / Name Resolution as required.

Session 14 & 15: (4T + 4L+ 5SL)

Theory:

- IT Act
- LDAP or Active Directory
- Introduction to Blockchain

Lab:

- Digitally Sign and Encrypt Email using the created certificate through Email Clients like Thunderbird or Windows Mail / Outlook.

Self-Learning: (5SL) (SL based on session 8-15)

- Explore about Aadhaar e-Sign ecosystem
- Importance of certificate chain trust
- Learn about how improper TLS configuration leads to MITM attacks
- How Strong authentication and MFA implementations impact security
- Write about the role of PKI in securing websites and email
- Study email encryption workflow using PGP / S-MIME
- List out the difference between OAuth vs OpenID vs Traditional Authentication
- Case Study: Digital signature dispute in court
 - ◆ Analyze about Legal validity of digital signatures
 - ◆ Role of trusted CA and timestamping